

Lesson 02: Classification of Differential Equations

Professor Dave Explains Differential Equations Master Series · Lesson 2

Video: Classification of Differential Equations Course: Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that classifying an equation is like diagnosing a medical condition: before you can prescribe the cure (an integration technique), you must precisely identify the disease. You must immediately recognize whether the system involves one independent variable (ODE) or multiple spatial dimensions (PDE), pinpoint the highest derivative present (order), and rigorously test whether the unknown function behaves in a strictly linear manner.

2. Core Theoretical Framework & Rigorous Formulations

Taxonomy Criteria

Property	Definition	Exam Diagnostic Rule
Type	ODE vs PDE	ODEs have 1 independent variable (d/dx); PDEs have 2+ independent variables ($\partial/\partial x, \partial/\partial t$).
Order	Highest derivative	Look for the highest derivative operator $y^{(n)}$; do not confuse with powers/exponents!
Linearity	$a_n(x)y^{(n)} + \dots + a_0(x)y = g(x)$	y and all derivatives must appear to power 1 only; no cross-products (yy'); no nonlinear functions ($\sin y, e^y$).

3. Fully Solved Whiteboard Problems

Concordia Diagnostic Drill: Full Classification Battery

Problem Statement: Classify the following equations by type, order, and linearity. If nonlinear, explicitly isolate the offending term: 1. $x^3y''' + 4xy' - 5y = \cos(x)$ 2. $\frac{d^2y}{dx^2} + y\frac{dy}{dx} = 0$ 3. $\frac{\partial^2u}{\partial x^2} + \frac{\partial^2u}{\partial y^2} = \rho(x, y)$ 4. $\left(\frac{dy}{dx}\right)^3 + y = x$

- Equation 1:** $x^3y''' + 4xy' - 5y = \cos(x)$

Type: ODE. Order: 3rd. Linearity: Linear (coefficients depend only on independent variable x ; y, y', y'' appear strictly to degree 1).
- Equation 2:** $y'' + yy' = 0$

Type: ODE. Order: 2nd. Linearity: Nonlinear. Offending term: $y \cdot y'$ (cross-multiplication of the dependent variable by its derivative).
- Equation 3:** $\partial_{xx}u + \partial_{yy}u = \rho(x, y)$

Type: PDE (independent variables x and y). Order: 2nd. Linearity: Linear (Poisson's equation).
- Equation 4:** $(y')^3 + y = x$

Type: ODE. Order: 1st (highest derivative is 1st order). Linearity: Nonlinear. Offending term: $(y')^3$ (first derivative raised to the 3rd power).

⚠ Common Exam Pitfall & Trapdoor Avoidance

Never confuse the order of a derivative with the power to which it is raised. $(y')^4$ is a first-order nonlinear term, while $y^{(4)}$ is a fourth-order linear term.

4. Key Takeaways & Exam Strategy Formula Card

🎯 Core Takeaway & Exam Rule

Order determines the required number of initial conditions; linearity determines whether superposition applies.

- Video reference: [Classification of Differential Equations](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.